Appendix A
Table Overview of the Included Articles with Their Concerns(Focus), Methods, Results, MOOC Weakness Themes

No	Author and Year	Title	Journal	Research Focus	Research Method	Results	Decision	Reason for Exclusion / Retrieval Status	MOOC Weakness Themes Coded (included studies only)
1	Onah, D.F.O., Pang, E.L.L., & Sinclair, J.E. (2021)	Investigating Self-Regulation in the Context of a Blended Learning Computing Course	Journal of Information Studies (University College London)	Investigating students' Self-Regulated Learning (SRL) capacity in the context of blended MOOC learning in a computing course, using the specially developed eLDa platform.	Quantitative – Action Research • Instrument: MOSLQ survey (adapted from OSLQ) with 5-point Likert scale • Sample: 27 students (survey 1); 17 respondents (survey 2) • Analysis: Descriptive and predictive statistics using SPSS • Setting: University of Warwick, UK	• Weakest SRL dimensions: Self-Evaluation (2.90) and Time Management (2.97) • Strongest SRL dimensions: Environment Structuring (3.78) and Goal Setting (3.60) • Significant individual variation (scores 2.43–4.14) • Students' academic achievement remained high (average 22.52/25) • Help-seeking and social interaction play an important role in blended learning • SRL is contextual and depends on the learning mode	Include	-	Self-directed learning difficulties
2	Gutiérrez-Santiuste, E., Gámiz-Sánchez, V.M., & Gutiérrez-Pérez, J. (2015)	MOOC & B-learning: Students' Barriers and Satisfaction in Formal and Non-formal Learning Environments	Journal of Interactive Online Learning Vol. 13, No. 3, Spring 2015 ISSN: 1541-4914	Comparing communication barriers and satisfaction levels of students in two virtual learning formats: b-learning (formal) and MOOC (non-formal) at a university in Spain.	Mixed Methods (Quantitative & Qualitative) • Instrument: Semi-structured questionnaire, 4-point Likert scale + open-ended questions • Sample: 249 students (64 formal / 185 non-formal) • Quantitative analysis: Mann-Whitney U test, Kolmogorov-Smirnov, Levene's test, Cohen's d • Qualitative analysis: Content analysis (NVivo v.8), interobserver agreement (ICC=0.66) • Design: Ex post facto descriptive-exploratory	• Technical barriers are the most dominant barrier in both groups • The non-formal (MOOC) group experienced more technical and sociological barriers • The formal (b-learning) group experienced more cognitive barriers • Student satisfaction was high in both groups, but higher in the formal group • Significant differences found in: planning, content, instructor, and community aspects • Learning community formed more strongly in the formal environment • Assessment was the main source of dissatisfaction in MOOC	Include	-	Lack of interaction; Technical barriers; Evaluation weaknesses; Difficult platform navigation
3	Pilli, O., Admiraal, W., & Salli, A. (2018)	MOOCs: Innovation or Stagnation?	Turkish Online Journal of Distance Education (TOJDE) Vol. 19, No. 3, July 2018 ISSN: 1302-6488	Examining MOOC's innovative position in higher education using SWOT analysis based on academic documents to answer the question: to what extent does MOOC bring innovation to higher education?	Document Analysis with SWOT Framework • Data sources: Books, research reports, journal articles, conference proceedings, blogs, discussion forums • Search: Leiden University database (EBSCO, ProQuest, SAGE, Science Direct, Wiley, etc.) + Google • Keywords: MOOC + innovation • Number of documents: 52 final documents (from 97 initial results, 45 eliminated) • Language: English-language documents only	• Strengths: accessibility, lifelong learning, online learning community, experimentation, institutional brand expansion • Weaknesses: high dropout rate, expensive infrastructure, weak pedagogy, inadequate assessment • Opportunities: expanded reach, collaborative learning, personalized learning, alternative education • Threats: sustainability, education quality, business model, student identity, unrecognized credentials, faculty resistance • MOOC has entered the Plateau of Productivity phase since 2015 • MOOC's true innovation lies in its large-scale open-access component	Exclude	Review, meta-analysis, conceptual, or SWOT-literature article, no primary data from MOOC participants	-
4	Ji, Z. (2016)	Application and Empirical Investigation of New MOOC Teaching System in Computer Application Course	International Journal of Emerging Technologies in Learning (iJET), Vol. 11, Issue 5	The effect of the MOOC multimedia system on learning in a computer application course at a vocational higher education institution	Experiment with control and experimental groups (n=60 students); survey questionnaire to teachers (n=7) and students (n=60)	The MOOC multimedia system increased students' enthusiasm and learning efficiency. 80% of students rated the system as easy to operate. The experimental group achieved higher scores. Teachers appreciated the time efficiency despite requiring adaptation. Weaknesses: teacher training is needed and system interactivity needs improvement.	Exclude	Empirical or design-oriented study that does not report MOOC weaknesses from the participants' perspective	-
5	Fisher, A., Anderson, G. B., Peng, R., & Leek, J. (2014)	A Randomized Trial in a Massive Online Open Course Shows People Don't Know What a Statistically Significant Relationship Looks Like, But They Can Learn	PeerJ, 2:e589	The ability of statistics MOOC participants to visually detect significant relationships from scatterplots	Randomized trial within a Coursera MOOC (n=2,039 students); scatterplot survey with various graphical presentation conditions; logistic regression analysis	Participants correctly identified significant relationships only 47.4% of the time (low sensitivity) and 74.6% for non-significant relationships (specificity). Adding a trend line increased the tendency to call a relationship significant regardless of the actual value. Accuracy improved on repeated trials, indicating that learning can occur through practice.	Include	-	Evaluation weaknesses

No	Author and Year	Title	Journal	Research Focus	Research Method	Results	Decision	Reason for Exclusion / Retrieval Status	MOOC Weakness Themes Coded (included studies only)
6	Stathakarou, N., Zary, N., & Kononowicz, A. A. (2015)	Evaluation of Three Educational Use Cases for Using Virtual Patients in Massive Open Online Courses (MOOCs): A Delphi Study	Bio-Algorithms and Med-Systems, Vol. 11, No. 2	Evaluation of three educational use cases for using Virtual Patients (VP) in MOOCs for health education: (A) collective decision-making evaluation, (B) cultural repurposing, (C) computational models in short cases	Two-round Delphi method with a panel of 6 experts from three stakeholder groups; open-ended questionnaire and ranking of use cases based on 4 aspects	Use case A (collective decision evaluation) received the highest priority for strengthening clinical reasoning and ease of implementation. Use case B (cultural repurposing) was rated as making the best use of MOOC's potential. Use case C (computational model) was most interesting to participants. Each use case excelled in a different aspect.	Include	-	Lack of interaction; Technical barriers; Inadequate media format; Difficult platform navigation
7	Maciejowska, I. (2024)	Implementation of the Course 'Good Chemistry: Methodological, Ethical and Social Implications' – A Case Study	Chemistry Teacher International, Vol. 6, No. 4	Implementation of the EuChemS platform-based MOOC 'Good Chemistry' at several European universities, particularly Jagiellonian University Krakow, in a blended learning format	Case study (descriptive and exploratory) with data triangulation; questionnaire survey of lecturers (n=5) and students (n=25); analysis of online discussion forum activity	The blended learning format effectively ensured all students completed the MOOC (100% completion rate). Most popular modules: Scientific Misconduct and Animal Experiments. Forum activity increased after face-to-face sessions. Students appreciated personal reflection but were accustomed to transmissive learning. Weakness: international participation was not yet optimal.	Include	-	Content too complex; Lack of interaction
8	Goldin, S., Kong, S. Y. J., Tokar, A., Utunen, H., Ndiaye, N., Bahl, J., Appuhamy, R., & Moen, A. (2021)	Learning From a Massive Open Online COVID-19 Vaccination Training Experience: Survey Study	JMIR Public Health and Surveillance, Vol. 7, No. 12	Evaluation of the learning experience and impact of COVID-19 vaccination training via the OpenWHO MOOC for health workers	Quantitative analysis of OpenWHO platform data (n=53,595 participants); participant feedback survey (n=2,019); pre- and post-quiz analysis; study period December 2020 – April 2021	Completion rate of 56%, far above the industry MOOC benchmark (5-10%). Average score increased from 76% (pre-quiz) to 85% (post-quiz). 98% of respondents felt more confident. 65.6% of participants preferred online training. Main barriers: internet connectivity and the need for additional vaccine-specific content.	Include	-	Content too complex; Lack of interaction; Technical barriers; Evaluation weaknesses; Language barriers
9	Findyartini, A., Greviana, N., Hanum, C., Husin, J. M., Sudarsono, N. C., Krisnamurti, D. G. B., & Rahadiani, P. (2021)	Supporting Newly Graduated Medical Doctors in Managing COVID-19: An Evaluation of a Massive Open Online Course in a Limited-Resource Setting	PLOS ONE, Vol. 16, No. 9	Analysis and evaluation of the MOOC (MTPC-I) for Indonesian intern doctors in managing COVID-19 in a limited-resource setting	Observational study with quantitative and qualitative approaches; Quality Reference Framework (QRF) analysis; pre- and post-test; satisfaction questionnaire (n=3,424 Batch 1, n=2,462 Batch 2); thematic analysis of qualitative data	Part A completion rate: 93.52% (Batch 1) and 90.55% (Batch 2). Post-test scores increased significantly across all topics ($p<0.005$). Participants appreciated the flexibility of access and variety of material. Main barriers: unstable internet connection in remote areas, shorter access duration than expected. Use of Indonesian language increased participation.	Include	-	Content too complex; Lack of interaction; Technical barriers; Inadequate media format; Evaluation weaknesses; Difficult platform navigation
10	Sharov, S., Biliatska, V., Kolmakova, V., Chernova, G., & Miroshnichenko, M. (2025)	Use of Massive Open Online Courses (MOOC) by Teachers of Ukrainian Universities: Survey Results	International Journal of Engineering Pedagogy (iJEP), Vol. 15, No. 3	Survey of MOOC use by Ukrainian university lecturers for self-development and the teaching process, and identification of MOOC's advantages and weaknesses	Google Forms questionnaire survey (n=2,824 lecturers from 155 universities); combination of closed and open-ended questions; hierarchical clustering for standardized response (SR) analysis; SQL database-based analysis	88.8% of lecturers use MOOCs. Most popular platforms: Prometheus (54.7%) and Coursera (31.2%). Main benefits: increased knowledge (18.1%), enrichment of practical experience (14.5%), and additional learning materials (12.9%). Weaknesses: poor connectivity (9.4%), technical problems (6.2%), low course quality (6.5%), and lack of teacher motivation (6.2%).	Include	-	Lack of interaction; Technical barriers; Inadequate media format
11	Draus, P. (2022)	Utilizing Student Views of Online, On-Ground, and Virtual Rotation Instructional Modalities for Future Development of Blended Learning Strategies	Issues in Information Systems, Vol. 23, No. 2	Student perceptions of three instructional modalities (face-to-face, online, virtual rotation) for future development of blended learning strategies, including MOOC integration	9-month qualitative online survey (n=72 students from 6 classes); thematic analysis with one round of theme reduction; open-ended questions about positive and negative aspects of each modality	50% of students preferred face-to-face, 38% preferred online. Advantages of face-to-face: direct feedback and interactivity. Advantages of online: flexibility and convenience. Virtual rotation received the most negative feedback (confusing, easily distracting). MOOC integration in blended learning can address online learning's weaknesses (slow feedback, social isolation).	Include	-	Lack of interaction; Self-directed learning difficulties
12	Negrin, M., Macerata, E., Concia, F., Mariani, M., Mossini, E., Das, S., et al. (2023)	A MOOC in Nuclear- and Radio-Chemistry: from the Design to the Feedback	Journal of Radioanalytical and Nuclear Chemistry Vol. 332, pp. 1549–1555	Development and evaluation of the MOOC "Essential Radiochemistry for Society" as a response to declining expertise and student interest in nuclear and radiochemistry (NRC) in Europe.	Case Study + Survey & Questionnaire Quantitative platform data analysis, user satisfaction survey, weekly questionnaire, and focus group with a pilot group of 50 students.	• The MOOC attracted 203 users from 20 countries in its pilot edition; 25% completed and earned a certificate. • Average final exam score was very high (0.91/1), indicating the effectiveness of the pedagogical framework. • Video-based modules, practical examples, and laboratory images produced higher scores. • Students found difficulty with complex concepts that were not sufficiently explained (Weeks 2 & 3). • The first annual edition was attended by 197 users from 29 countries; the second edition showed stable participation. • Effective promotion to teachers was a key factor in outreach success.	Include	-	Content too complex; Technical barriers; Inadequate media format

No	Author and Year	Title	Journal	Research Focus	Research Method	Results	Decision	Reason for Exclusion / Retrieval Status	MOOC Weakness Themes Coded (included studies only)
13	Vista, A., Care, E., & Griffin, P. (2015)	A New Approach Towards Marking Large-Scale Complex Assessments: Developing a Distributed Marking System that Uses an Automatically Scaffolding and Rubric-Targeted Interface for Guided Peer-Review	Assessing Writing Vol. 24, pp. 1–15	Development of a scaffolded peer-review system (ARISE), based on automatic scaffolding and a rubric-targeted interface, to address the limitations of automated assessment and peer-grading reliability in large classes and MOOC contexts.	System Design & Development Design and implementation of a web-based API (ARISE); descriptive conceptual approach; plans for a 2×3 factorial experimental validation in a later stage.	• The ARISE system was successfully developed with structural scaffolding (readability, keywords) and relational scaffolding (higher-order thinking indicators). • The system automatically highlights content relevant to the instructor's rubric, helping peer-reviewers focus. • Differs from Automated Essay Scoring because final grading is still done by humans. • Guided scaffolding has the potential to improve inter-rater reliability equivalent to pre-assessment training. • The system can be applied across disciplines and supports collaborative learning and self-assessment.	Exclude	Empirical or design-oriented study that does not report MOOC weaknesses from the participants' perspective	-
14	He, W. (2022) ⚠ RETRACTED – August 2023	Assessing the Impact of the MOOC Learning Platform on the Comprehensive Development of English Teachers at College Level under "Double First-Rate" by Utilization of the SWOT Analysis in Hunan Province, China	Journal of Advanced Transportation Vol. 2022, Art. ID 9202921 (RETRACTED)	Examining the impact of the MOOC platform on the professional development of college-level English teachers within the context of the "Double First-Rate" policy in China, using SWOT analysis.	Quantitative (Status: Questionable) Time-series-based recommendation algorithm; Structural Equation Model (SEM); SWOT Analysis. ⚠ Article retracted due to systematic manipulation of the publication process.	WARNING: This article was RETRACTED by Hindawi/Wiley (August 2023) due to: • Inconsistencies in research scope and description • Inconsistent data availability • Improper citations & irrelevant content • Peer-review manipulation Findings cannot be relied upon. Not recommended for use in the SLR.	Exclude	Article retracted by the publisher	-
15	Papadimitriou, A. (2024)	Theories, Strategies and Elements of Gamified MOOCs: A Systematic Literature Review	Asia Pacific Journal of Information Systems Vol. 34, No. 1, pp. 248–291	A comprehensive systematic literature review (SLR) of theories, strategies, and elements of gamification in gamified and adaptive gamified MOOCs to address high dropout rates, low engagement, and low completion rates.	Systematic Literature Review (SLR) Search of Scopus & Google Scholar; PICOS framework; PRISMA 2020 flow diagram; analysis of 143 articles (2011–2023); Cochrane RoB 2 bias evaluation.	• Most influential theories: Self-Determination Theory, followed by Flow Theory, Technology Acceptance Model, and Goal-Setting Theory. • Most influential strategies: theory-based (multi-component), challenge-based, team-based, persuasive-based, and story-based. • Most popular elements: points, badges, leaderboards, rewards, competition, and narrative. • Gamification has been proven to improve completion rate, motivation, engagement, interaction, and collaboration among students. • Adaptive gamification produces longer learning time compared to no adaptation. • Nearly half of the articles were published in 2019–2022, indicating rapid research growth.	Exclude	Review, meta-analysis, conceptual, or SWOT-literature article, no primary data from MOOC participants	-
16	Klemke, R., Antonaci, A., & Limbu, B. (2020)	Designing and Implementing Gamification: GaDeP, Gamifire, and Applied Case Studies	International Journal of Serious Games Vol. 7, No. 3, pp. 97–119	Development and validation of the Gamification Design Process (GaDeP) as a systematic framework for meaningful gamification design, as well as Gamifire as a platform-independent and scalable infrastructure for MOOC gamification.	Design & Development + Case Study Software engineering; 4 Gamifire case studies on different MOOCs; observational study of 15 master's students applying GaDeP over a 6-week course.	• GaDeP consists of 6 iterative phases: (1) scenario analysis, (2) problem definition, (3) theoretical framework, (4) game element selection, (5) design & implementation, (6) evaluation. • Gamifire was successfully developed as a cloud platform based on Google App Engine that is platform-independent and scalable. • Gamifire supports: stimulated planning, clans/teams, avatars, narrative, challenges, HUD, and communication. • In a cryptography MOOC, Gamifire's social elements increased social presence and sense of community. • GaDeP was successfully applied by students without a gamification background in 4 scenarios outside of MOOCs. • Main tension: platform-independence vs. meaningful gamification, resolved by mirroring content structure on the front-end.	Exclude	Empirical or design-oriented study that does not report MOOC weaknesses from the participants' perspective	-
17	Wan Nurhayati Wan Ab Rahman, Hazura Zulzalil, Iskandar Ishak, Ahmad Wiraputra Selamat (2020)	Quality Model for Massive Open Online Course (MOOC) Web Content	International Journal on Advanced Science Engineering Information Technology (IJASEIT)	Proposing a quality model for MOOC web content from the content provider's perspective, identifying relevant quality dimensions and quality factors for developing high-quality MOOC content.	Systematic Literature Review (SLR) with a Factor-Criteria-Metrics (FCM) approach; collection of factors from web content, e-learning, and MOOC literature (2010–2018); factor combination and categorization process; adaptation of the 7C framework and PDCA cycle.	A hierarchical MOOC web content quality model was proposed, consisting of 54 quality factors grouped into 10 categories (Conceptual, Massiveness, Openness, Video Quality, Usability, Engagement, Maintainability, Portability, Continuity) based on the Plan-Develop-Check-Live development phases. This model is the first to be developed from a content provider's perspective and is equipped with operational definitions for each factor.	Exclude	Review, meta-analysis, conceptual, or SWOT-literature article, no primary data from MOOC participants	-

No	Author and Year	Title	Journal	Research Focus	Research Method	Results	Decision	Reason for Exclusion / Retrieval Status	MOOC Weakness Themes Coded (included studies only)
18	Sadra Ahmadi, Zahra Nourmohamadzadeh, Babak Amiri (2023)	A hybrid DEMATEL and social network analysis model to identify factors affecting learners' satisfaction with MOOCs	Heliyon (Elsevier)	Identifying factors that influence MOOC user satisfaction based on analysis of user comments on the Coursera platform, and prioritizing these factors for platform improvement.	Hybrid approach: (1) Text mining using sentiment analysis (TF-IDF + SVM) and topic modeling (LDA/Latent Dirichlet Allocation) on ~12,000 comments from 100 Coursera courses; (2) Creation of a factor relationship matrix; (3) DEMATEL method to determine influence between factors; (4) Social Network Analysis (SNA) using Gephi software to analyze factor centrality; (5) Management matrix to prioritize factors.	62 factors influencing user satisfaction were identified, grouped into 5 categories: course content, instructor, course structure, course effects, and course support. The most influential factors were: clear and complete content (S1), easy exercises (S4), well-explained material (S13), short and quality videos (S20), and suitability for self-directed learning (S23). Main weaknesses: lack of exercises (W4), missing data on the final quiz (W7), and poor instructors (W17).	Exclude	Empirical or design-oriented study that does not report MOOC weaknesses from the participants' perspective	-
19	Nour Awni Albelbisi, Farrah Dina Yusop (2020)	Systematic Review of A Nationwide MOOC Initiative In Malaysian Higher Education System	The Electronic Journal of e-Learning (EJEL)	Analyzing and synthesizing research related to the national MOOC initiative in the Malaysian higher education system (2014–2018), providing an overview of MOOC growth in Malaysia, and identifying issues in national MOOC implementation.	Descriptive Systematic Literature Review (SLR) using systematic content analysis; literature search in Web of Science, Scopus, and Google Scholar with 8 main keywords; 5-stage Khan et al. (2003) methodology; PRISMA technique for article selection flow diagram; ultimately 25 studies analyzed and categorized into 3 sections.	Malaysia's MOOC was launched nationally in September 2014, hosted on the OpenLearning platform with more than 241 courses and 238,267 enrollments. MOOC research in Malaysia focused on two themes: (1) MOOC acceptance by students and lecturers (positive, as a lifelong learning tool), and (2) aspects of MOOC delivery. Five main issues: lack of course massiveness, most participants being Malaysian, low lecturer self-efficacy, limited internet access, and student anxiety in using MOOCs.	Exclude	Review, meta-analysis, conceptual, or SWOT-literature article, no primary data from MOOC participants	-
20	Rachel C. Kearney, Sundaralingam Premaraj, Becky M. Smith (Viewpoint 1); Gregory W. Olson, Anne E. Williamson, Georgios Romanos (Viewpoint 2) (2016)	Massive Open Online Courses in Dental Education: Two Viewpoints	Journal of Dental Education	Discussing the strengths and weaknesses of integrating MOOCs into dental education through two opposing viewpoints: whether MOOCs can significantly transform the traditional dental education curriculum.	Point/Counterpoint article (argument vs. counter-argument) based on literature review and personal observation; does not use empirical methods; relies on evidence from preliminary studies, meta-analysis (US Dept. of Education), and institutional experience.	Viewpoint 1 (Pro-MOOC): MOOCs have the potential to transform dental education, especially for basic science instruction, addressing faculty shortages, supporting blended learning (flipped classroom), and generating revenue from continuing education. Viewpoint 2 (Anti-MOOC): MOOCs are not yet ready for the dental curriculum due to: high dropout rate (5–20%), lack of meaningful faculty-student interaction, inability to replace clinical experience, accreditation and assessment challenges, and risk of academic dishonesty.	Exclude	Review, meta-analysis, conceptual, or SWOT-literature article, no primary data from MOOC participants	-
21	Azer Nouira, Lilia Cheniti-Belcadhi, Rafik Braham (2019)	An ontology-based framework of assessment analytics for massive learning	Computers & Applications in Engineering Education (Wiley)	Exploring existing learning analytics data models, identifying weaknesses in the xAPI specification in supporting assessment analytics, and proposing an enhanced ontological model to effectively support assessment analytics in MOOC environments.	Comparative analysis of 7 learning analytics data models (CAM, Activity Streams, Learning Registry Paradata, NSDL Paradata, IMS Caliper, LCDM, xAPI); development of an ontological model using Protégé; UNIX-based preprocessing (GREP, AWK, CUT, sort); extraction of assessment data from real MOOC log files (ICOO MOOC, Sousse University, >750,000 learning traces); development of a Java Semantic Web application using the JENA library for conversion to OWL.	Of the 7 learning analytics data models evaluated, only xAPI was able to support assessment analytics, but it has weaknesses: (1) inadequate assessment result information, and (2) no assessment context information. The proposed ontological model enriches xAPI with: additional metadata (number of correct/incorrect answers, unanswered questions, number of attempts) and assessment context (type, format, assessment technique). Of the ICOO MOOC with 3,470 enrollees, only 10% completed the assessment process, indicating a dropout rate of ~90%.	Exclude	Empirical or design-oriented study that does not report MOOC weaknesses from the participants' perspective	-
22	Stefanie Schütte, Sophie-Hélène Goulet-Ebongue, Khamsa Habouchi (2018)	Could massive open online courses improve health and medical education?	Public Health Forum (De Gruyter)	Discussing the strengths, weaknesses, opportunities, and threats (SWOT) of MOOCs as a health and medical education tool, with a specific focus on low- and middle-income countries (LMICs).	SWOT Analysis (Strengths, Weaknesses, Opportunities, Threats) based on literature review and institutional experience of the Centre Virchow-Villermé (Paris); does not use quantitative empirical methods; descriptive-qualitative analysis.	Strengths of MOOCs in health education: verified certification, flexibility, cross-border geographic access, improved public health literacy, and direct feedback. Weaknesses: reaches only certain populations, lack of physical interaction, high dropout rate, language and cultural barriers. Opportunities: empowerment of health workers and patients, bridging knowledge gaps, democratization of education. Threats: worsening health disparities between developed countries and LMICs. MOOCs play an important role in health crisis responses such as the Ebola outbreak. Centre Virchow-Villermé has produced 20 public/global health MOOCs since 2013.	Exclude	Review, meta-analysis, conceptual, or SWOT-literature article, no primary data from MOOC participants	-

No	Author and Year	Title	Journal	Research Focus	Research Method	Results	Decision	Reason for Exclusion / Retrieval Status	MOOC Weakness Themes Coded (included studies only)
23	Kerry Wu (2013)	Academic libraries in the age of MOOCs	Reference Services Review (Emerald)	Reporting on recent MOOC developments, exploring the strengths and weaknesses of MOOCs, and reflecting on the possible relationship between academic libraries and MOOCs, and the role libraries can play.	Mixed literature review based on current news and the author's personal observations as a MOOC student; a conceptual paper; does not use empirical methods or primary data collection.	MOOCs create a global learning community that benefits students and universities. MOOCs excel in content delivery but are weak in interaction, mentoring, and assessment. The role of academic libraries in the MOOC ecosystem includes: (1) copyright permissions and alternative open-access search, (2) course production, (3) policy and best practice development, (4) information literacy module development, and (5) collaboration as an embedded librarian. The concept of 'iTunization of information' is introduced as a potential new business model in which academic content is sold individually to MOOC students. Libraries need to be actively involved in discussions about MOOCs at their respective institutions.	Exclude	Review, meta-analysis, conceptual, or SWOT-literature article, no primary data from MOOC participants	-
24	Yang, Guojing (2025)	Enhanced IVIFN–ExpTODIM–MABAC Technique for Multi-attribute Group Decision-Making and Applications to College English Teaching Quality Evaluation Under Interval-Valued Intuitionistic Fuzzy Sets	International Journal of Fuzzy Systems	Evaluation of college English teaching quality based on MOOCs involving Multi-Attribute Group Decision-Making (MAGDM)	Evaluation	-	Not retrieved	Full text not retrieved (restricted access) excluded at retrieval stage, not assessed for eligibility	-
25	Wang, Lianhong, Li, Xiaoyao, Luo, Zhihui, Hu, Zinan, Yan, Qing (2024)	Multivariate Cognitive Response Framework for Student Performance Prediction on MOOC	IEEE Transactions on Knowledge and Data Engineering	A cognitive diagnostic model (CDM) can reveal the potential relationship between students' knowledge levels, exam question characteristics, and corresponding question scores, as well as predict students' future performance.	Experiment collecting student learning activity logs to calculate corresponding effort characteristics.	Experimental results on simulation studies and real data application on MOOC show that MvCRF achieves better predictive performance, robustness, and interpretability than the baseline CDM	Exclude	Empirical or design-oriented study that does not report MOOC weaknesses from the participants' perspective	-
26	Xiong, Qian Li (2018)	College english MOOC teaching on SWOT analysis	Kuram ve Uygulamada Egitim Bilimleri	This paper also examines the strengths, weaknesses, and opportunities of MOOC teaching, and suggests that MOOC teaching be combined with traditional teaching	-	-	Not retrieved	Full text not retrieved (restricted access) excluded at retrieval stage, not assessed for eligibility	-